MILITARY SPECIFICATION

PROBE, RADIAC DT-590( )/PDR-56F

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Probe, Radiac DT-590/PDR-56F (X-ray probe), part of alpha survey meter Radiac Set AN/PDR-56F.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefore in the cited specifications or (ii) issues different from those specified in the cited specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of
MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	-	Marking of Electronic Items
MIL-F-14072	-	Finishes for Ground Electronic Equipment
MIL-P-15024	-	Plate, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-E-55585	-	Electronic Equipment and Parts, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6665

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461A	- Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-726	- Packaging Requirement Codes
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810	- Environmental Test Methods

DRAWINGS

DL-5072047	- Probe, Radiac DT-590( )/PDR-56F

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.2g).

3.2 Construction. The X-ray Probe shall be constructed in accordance with the requirements of this specification and of Data List DL-5072047.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:04Z
Check the source to verify that this is the currentversion before use.

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.4 Finish. All surfaces of the X-ray probe, requiring a protective coating, shall be finished in accordance with MIL-F-14072 (see 4.4).

3.5 Nameplate and product marking.

3.5.1 General. Marking shall be in accordance with MIL-M-13231 (see 4.4).

3.5.2 Nameplates. Each X-ray probe shall have a nameplate conforming to MIL-P-15024/6.

3.5.3 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.4 Serial numbers. Each X-ray probe shall be serialized (see 4.4). The serial number shall be identical to the matching AN/PDR-56F radiac set.

3.6 Interchangeability. Like units, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.6).

3.7 Burn-in. Each X-ray probe shall be subjected to a burn-in of not less than 24 hours, the last 12 of which shall be failure free (see 4.7).

3.8 Weight. The maximum total weight of the X-ray probe shall not exceed 2-1/2 pounds (see 4.8)

3.9 Systems safety engineering.

3.9.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.9).

3.9.2 Corner and edge rounding. All exposed corners shall be rounded to eliminate possible injury to personnel due to lacerations and cuts.

3.9.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, surge arrestors and lenses).

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:04Z
Check the source to verify that this is the current version before use.

3.10 Performance requirements.

3.10.1 Energy range. The X-ray probe shall selectively measure X-rays in the energy range of 14 to 21 Kev (see 4.10.1).

3.10.2 Sensitivity ranges. Sensitivity ranges of the X-ray probe with AN/PDR-56F radiacmeter shall be equivalent to 0.1 to 200 milligrams per square meter (mg/m^2) of Pu-239. Ranges of indication shown on the AN/PDR-56F radiacmeter shall be as follows: (see 4.10.2).

Pu-239 concentration (mg/m^2)	AN/PDR-56F scale ranges (cpm)
0.1 - 0.2	0 - 1K
0 - 2.0	0 - 10K
0 - 20	0 - 100K
0 - 200	0 - 1M

3.10.3 Accuracy. (see 4.10.3)

3.10.3.1 Conditions. The following conditions shall apply with respect to accuracy requirements of the X-ray probe:

a. Thin, uniformly distributed Pu-239 sources with size and alpha energy characteristics equivalent to those of the AN/UDM-7B alpha calibrator shall be used.

b. The source-to-X-ray probe geometry shall be the same as that used for calibration (see 3.10.4) of the X-ray probe which is approximately one to two inches.

c. Background readings shall be subtracted from the actual readings taken of the Pu-239 source.

3.10.3.2 Response. Response of the X-ray probe shall be linear. Radiacmeter reading at any point over the upper 50 percent of the 0-1K counts per minute (cpm) scale and the upper 90 percent of all other ranges shall be within $\pm$30 percent of the readings required by the actual Pu-239 concentrations based on linear interpolations of the full scale relationships of 3.10.2 (see 4.10.3.1).

3.10.3.3 Background response. The X-ray probe response to normal background radiation, PM tube noise, bright sunlight and probe circuit noise shall be less than 250 cpm indicated on the 1K scale of the AN/PDR-56F radiacmeter (see 4.10.3.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

3.10.3.4 Gamma response. Meter indications on the AN/PDR-56F radiacmeter with the X-ray probe attached, shall be less than 40K cpm when the probe is placed in a 10 milli-roentgens per hour (mr/hr) cobalt-60 gamma field (see 4.10.3.3).

3.10.4 Calibration. The X-ray probe shall be calibrated using the AN/UDM-7B alpha calibration source (Pu-239). Before calibrating the X-ray probe, the AN/PDR-56F Radiacmeter, with the alpha probe attached, shall be calibrated following the procedures described in 4.10.4. The X-ray probe shall then be attached to the AN/PDR-56F and calibration of the X-ray probe shall be accomplished without readjusting the AN/PDR-56F calibration controls or the high voltage setting of the radiacmeter (see 4.10.4).

3.10.5 Reproducibility. Any meter reading on that portion of the scale range on which accuracy requirements apply (see 3.10.3) shall be reproducible to within 15 percent when the probe is placed in an identical field under identical operational and environmental conditions (see 4.10.5).

3.10.6 Battery life. The AN/PDR-56F radiacmeter with the X-ray probe attached shall operate as specified for at least 40 hours per set of batteries, including 4 hours intermittent operation of the meter light. Changing of the batteries in the radiacmeter shall not require recalibration of the X-ray probe (see 4.10.6).

3.10.7 System check. Operation of the X-ray probe in the presence of a radiation field shall be checked electronically (see 4.10.7).

3.10.8 Light tightness. The X-ray probe shall be subjected to a light intensity of 10,000 foot candles on the surface of the crystal area. The radiacmeter shall indicate no more than 250 cpm (see 4.10.8).

3.11 Service conditions. The X-ray probe shall be subjected to and meet the service conditions below. For the purpose of service conditions, specified performance is defined as meeting the following requirements:

Reqt. para.	Test para.	Measurement
3.10.1	4.10.1	Energy range
3.10.2	4.10.2	Sensitivity ranges
3.10.3	4.10.3	Accuracy
3.10.4	4.10.4	Calibration
3.10.5	4.10.5	Reproducibility
3.10.7	4.10.7	System check

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:04Z
Check the source to verify that this is the current version before use.

3.11.1 High temperature. The X-ray probe shall meet the performance requirements of 3.10 at 125°F (52°C). The probe shall meet specified performance without degradation after a 30 minute stabilization period at room ambient temperature (see 4.11.1).

3.11.2 Low temperature. The X-ray probe shall meet the accuracy requirement of 3.10.3 at -25°F (-32°C). The probe shall meet specified performance without degradation after a 30 minute stabilization period at room ambient temperature (see 4.11.2).

3.11.3 Humidity. The X-ray probe shall exhibit no physical damage such as corrosion, rust blistering, swelling, or deterioration of parts and materials and shall meet the accuracy requirement of 3.10.3 when subjected to 94 ± 4 percent relative humidity. The probe shall meet specified performance without degradation after completion of the test (see 4.11.3).

3.11.4 Altitude. The X-ray probe shall meet the accuracy requirement of 3.10.3 at 10,000 ft (3048 m) above sea level. The probe shall meet specified performance without degradation after completion of the test (see 4.11.4).

3.11.5 Splashproof. The X-ray probe shall be splashproof when tested in accordance with Table II of MIL-STD-108 (see 4.11.5).

3.11.6 Vibration. The X-ray probe shall be operated during the Type I vibration test of MIL-STD-167. There shall be no evidence of collision between parts of the equipment as a result of the test. The X-ray probe shall meet specified performance without degradation after completion of the test (see 4.11.6).

3.11.7 Shock, bench handling. The X-ray probe shall operate without degradation in specified performance and shall suffer no mechanical damage after being subjected to the bench handling test specified in 4.11.7.

3.12 Electromagnetic interference (EMI). The X-ray probe shall meet the following requirements of MIL-STD-461A:

RE02
RS03

For RS03, the performance of the X-ray probe shall comply with the requirements of 3.10 when immersed in electric fields of 1.0 volt/meter having the following modulation characteristics:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

Frequency	Modulation Characteristics
14 KHz to 30 MHz	AM, 50 percent, 1 KHz tone
30 MHz to 76 MHz	FM, $\pm$ 8 KHz deviation, 1 KHz tone
76 MHz to 400 MHz	AM, 50 percent, 1 KHz tone
400 MHz to 10 GHz	Pulse, 0.1 us pulse width, 400 pps

3.13 Reliability. The X-ray probe shall have a specified (θ_o) MTBF of 1000 hours when operating under the following conditions (see 4.13):

a. Ambient temperature 68°F to 122°F (20°C to 50°C)
b. Ground mobile environment
c. 100 percent duty cycle

3.14 Government loaned property. Unless otherwise specified by the contract or purchase order, the government will provide on a loan basis, the following items for use in the production of this equipment:

Radiac Set	AN/PDR-56F
Calibrator Set, Radiac	AN/UDM-7B (alpha source)

3.15 Workmanship. The X-ray probe shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268 (see 4.4):

Plastic materials and parts	Welding
Controls	Wiring practices, internal, including:
Screws, other threaded devices, and related parts	Cabling
	Clearance
Brazing	Protection
Cleaning	Splicing and stretching
Riveting	Connections, general
Soldering	Grounding

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.15).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish nine first article units of the X-ray probe for Group A, B, C, and D testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in Table I, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test report shall be as required in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

Inspections	Reqt Para	Test Para	Order of tests: Units 1, 2	Units 3, 4	Units 5, 6	Units 7-9
1. Inspections covered by subsidiary documents. 1/	3.3 3.4 3.5 3.15	4.4	Inspection to be performed on all units			
2. Burn-in 1/	3.7	4.7	Tests to be performed on all units			
3. Group A inspection 1/	See Table II		Tests to be performed on all units			
4. Group B inspection 1/	See Table III		Tests to be performed on all units			
5. Group C inspection						
Battery life	3.10.6	4.10.6	1			
Altitude	3.11.4	4.11.4	4			
High temperature	3.11.1	4.11.1		2		
Low temperature	3.11.2	4.11.2		1		
Humidity	3.11.3	4.11.3		3		
Vibration	3.11.6	4.11.6			1	
Shock, bench handling	3.11.7	4.11.7			2	
Splash proof	3.11.5	4.11.5			3	
EMI	3.12	4.12	3			
Weight	3.8	4.8	2			
6. Group D inspection (Reliability)	3.13	4.13				1

1/ Inspections 1 to 4 shall be performed in the order shown before subjecting the equipments under test to any other inspection requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:04Z
Check the source to verify that this is the current version before use.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Parts, materials, and processes; general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.15

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4 and 4.7. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspection in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE II. Group A inspection.

Inspection	Reqt Para	Test Para	AQL Major	AQL Minor
Visual & mechanical	3.15	4.14	1%	4%
Light tightness	3.10.8	4.10.8	1%	4%
Electrical				1/
Sensitivity	3.10.2	4.10.2	1%	
Energy range	3.10.1	4.10.1	1%	
Calibration	3.10.4	4.10.4	1%	

1/ All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 20260-22T20:04Z
Check the source to verify that this is the current version before use.

4.5.2.1 Group B sampling plans. The Group B sample plans, for the AQL's listed in Table III, shall be as follows:

AQL	Inspection level
4 per cent	S-4
6.5 per cent	S-4

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE III. Group B inspection.

Inspection	Reqt Para	Test Para	AQL
Accuracy	3.10.3	4.10.3	4%
Reproducibility	3.10.5	4.10.5	6.5%
System check	3.10.7	4.10.7	4%
Interchangeability	3.6	4.6	6.5%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. Six (6) samples of X-ray probe shall be selected at random, two (2) samples for each subgroup in Table IV from every 300 units or fraction thereof greater or equal to 150 units. If the fraction is less than 150, the selection of Group C samples for the preceding test shall include units from the remaining fraction.

4.5.3.2 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

TABLE IV. Group C inspection.

Inspection	Reqt Para	Test Para
Subgroup 1		
Weight	3.8	4.8
Electromagnetic interference 1/	3.12	4.12
Altitude	3.11.4	4.11.4
Battery life	3.10.6	4.10.6
Subgroup 2		
High temperature	3.11.1	4.11.1
Low temperature	3.11.2	4.11.2
Humidity	3.11.3	4.11.3
Subgroup 3		
Vibration	3.11.6	4.11.6
Shock, bench handling	3.11.7	4.11.7
Splashproof	3.11.5	4.11.5

1/ To be performed on the first production lot only.

4.5.4 Group D inspection. This inspection shall consist of the test specified in Table V and shall be performed on units which have been subjected to and have passed Group A and Group B inspections.

4.5.4.1 Sampling for Group D inspection. Three (3) samples of the X-ray probe shall be selected at random from every five hundred (500) units or fraction thereof greater or equal to two hundred fifty (250) units. If the fraction is less than two hundred fifty (250), the selection of Group D samples for the preceding test shall include units from the remaining fraction.

Table V. Group D inspection.

Inspection	Reqt Para	Test Para
Reliability	3.13	4.13

4.5.4.2 Group D failures. In case of Group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:04Z
Check the source to verify that this is the current version before use.

4.5.4.3 Reinspection of Group D sample units. Unless otherwise specified, sample units which have been subjected to, and have passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

d. Size and form of special threads.

4.7 Burn-in.

4.7.1 Testing. Each X-ray probe shall be subjected to a 24 hour burn-in, the last 12 hours of which shall be failure free. The conditions for the burn-in shall be the same as for the reliability test. The failures shall be recorded and analyzed but shall not be considered relevant for computation of MTBF (see 3.7).

4.7.2 Compliance. Units passing burn-in shall be submitted for conformance testing without benefit of repair, cleaning, realignment or other processing. If, after completion of burn-in, the unit fails conformance testing, it shall be reworked and resubmitted to the final 12 hour period of burn-in.

4.8 Weight. The X-ray probe shall be weighed. The weight shall not exceed that given in 3.8.

4.9 System safety engineering inspection. A visual inspection shall be performed in the presence of a government safety representative to determine that all requirements of 3.9 have been incorporated in the equipment. Inability to meet these requirements shall constitute a failure of the test. Results of this inspection shall be included in the Safety Assessment Report, if required in the contract.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:04Z
Check the source to verify that this is the current version before use.

4.10 Performance tests.

4.10.1 Energy range. With the X-ray probe in the operating mode and properly calibrated, the single channel analyzer shall selectively measure X-rays in the energy range of 14 to 21 Kev. Monoenergetic X-ray sources shall be used in this test (see 3.10.1).

4.10.2 Sensitivity ranges. With the X-ray probe in the operating mode, irradiate the probe with source A and C respectively of the alpha source AN/UDM-7B. The source to probe distance shall be approximately one (1) inch. The read-out of the AN/PDR-56F radiacmeter shall be as specified in 3.10.2.

4.10.3 Accuracy. With the X-ray probe in the operating mode, irradiate the probe with sources A and C respectively of the alpha source AN/UDM-7B. The readout shall be within $\pm$ 20 percent of the actual Pu-239 concentration. The conditions specified in section 3.10.3.1 shall be followed in the test.

4.10.3.1 Response. With the X-ray probe in the operating mode, irradiate the probe with sources A and C respectively of the alpha source AN/UDM-7B. Response of the X-ray probe shall be linear as specified in 3.10.3.2.

4.10.3.2 Background response. With the X-ray probe in the operating mode, the readout due to normal background radiation shall be less than 250 cpm (see 3.10.3.3).

4.10.3.3 Gamma response. With the X-ray probe in the operating mode, irradiate the probe with a Co-60 gamma field of 10 mR/hr. The readout shall be less than 40K cpm (see 3.10.3.4).

4.10.4 Calibration. With the X-ray probe in the operating mode, the probe shall be calibrated using AN/UDM-7B alpha source A. The calibration of the probe shall be indicated in observed countrate (counts per minute) per mg/m^2 of Pu-239 concentration. A mono-energetic (approximately 17 kev) X-ray source shall be used for adjusting the gain of the X-ray probe amplifier. The 14 to 21 kev Pu-239 X-ray peak shall be centered within the single channel analyzer (see 3.10.4).

4.10.5 Reproducibility. With the X-ray probe in the operating mode, the probe shall be irradiated with Pu-239 sources of equal concentrations. The readouts shall be within $\pm$ 15 percent excluding the calibration error of the Pu-239 sources (see 3.10.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.10.6 Battery life. With the X-ray probe in the operating mode, the probe shall be irradiated with Pu-239, source A of AN/UDM-7B. The readout shall be within ± 20 percent over the entire battery life of 40 hours per set of batteries, including 4 hours intermittent operation of the AN/PDR-56F radiacmeter light (see 3.10.6).

4.10.7 System check. With the X-ray probe in the operating mode, irradiate the probe with the check source attached on the back of the AN/PDR-56F radiacmeter. The readout shall be in the upper 50 percent of the 10K scale (see 3.10.7).

4.10.8 Light tightness. With the X-ray probe in the operating mode, subject the surface of the crystal area to a light intensity of 10,000 foot candles. The radiacmeter reading shall be no more than 250 cpm (see 3.10.8).

4.11 Service conditions. The service condition tests shall be performed as detailed below. Each X-ray probe subjected to these tests shall have passed Group A and B tests and shall be resubmitted to and pass the Group A tests after completion of all service condition tests. Where operation of the test item is required during any service condition test, the test item shall be required to pass the Accuracy test of 4.10.3.

4.11.1 High temperature test. The X-ray probe shall be subjected to the test of Method 501.1, Procedure I of MIL-STD-810. The operating temperature in Step 4 shall be 120° F (49° C) for three hours followed by one hour at 125° F (52° C) (see 3.11.1).

4.11.2 Low temperature test. The X-ray probe shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -70° F (-57°C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (Step 4) shall be -25° F (-32° C) (see 3.11.2).

4.11.3 Humidity test. The X-ray probe shall be subjected to the test of Method 507.1, Procedure IV, of MIL-STD-810. In Step 5 the high temperature shall be 122° F (50° C) (see 3.11.3).

4.11.4 Altitude test. The X-ray probe shall be subjected to the test of Method 500.1, Procedure I of MIL-STD-810. The equipment shall be held at 40,000 feet (12,192m) non-operating for one (1) hour. The operating altitude in Step 2 shall be 10,000 feet (3,048 m) (see 3.11.4).

4.11.5 Splashproof test. The X-ray probe shall be tested for compliance with the requirements of 3.11.5 using the approved test methods of MIL-STD-108 (see 3.11.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.11.6 Vibration test. The X-ray probe shall be tested for compliance with the requirement of 3.11.6 when tested in accordance with Type I vibration test of MIL-STD-167.

4.11.7 Shock, bench handling; test. The X-ray probe shall be subjected to the test of Method 516.2, Procedure V of MIL-STD-810 (see 3.11.7).

4.12 Electromagnetic interference test. The X-ray probe shall be tested to determine compliance with the requirements of 3.12. Test methods and procedures shall be in accordance with MIL-STD-462. Testing shall be performed with the probe terminated into the AN/PDR-56F.

4.13 Reliability testing.

4.13.1 First article and Group D inspection. To determine compliance with 3.13 during first article and Group D inspections, the equipment shall be tested in accordance with Test Plan VC of MIL-STD-781. The specified MTBF for all reliability testing shall be 1000 hours. The lower test MTBF (θ_1) shall be 333 hours.

4.13.2 Test conditions. The environment for first article reliability inspection shall be under the following conditions:

a. Temperature and on-off cycle. The item shall be continuously operated. Temperature cycling shall be accomplished by alternately elevating the chamber temperature to and maintaining it at 122°F (50°C) for 4 hours after equipment stabilization, then reducing the chamber temperature to 73°F (23°C) for 4 hours after equipment stabilization and then cooling to -4°F (-20°C) and maintaining that temperature for 4 hours after equipment stabilization. This cycle shall continue for the duration of the test. The average range of temperature change shall be 9°F (5°C) per minute.

b. Vibration. For at least 10 minutes out of each hour during the test, the equipment shall be subject to vibration at a non-resonant frequency between 20 and 60 Hz. The peak magnitude of the vibration when measured at the mounting point of the equipment shall be $2.2g \pm .22g$.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.13.3 Performance parameters. The following performance parameters shall be measured and recorded at least daily during reliability testing:

a. Accuracy (see 3.10.3)

b. Calibration (see 3.10.4)

c. Reproducibility (see 3.10.5)

d. System check (see 3.10.7)

4.13.4 Failure definitions. A failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes departure or deviation from performance parameters required by this specification. Failure confirmation categorizing shall be in accordance with MIL-STD-781. Proper instrumentation shall be provided to insure immediate recognition of failure.

4.13.5 Minimum sample test times. No single equipment on time shall be less than one half the average operating time of all equipment in each respective test.

4.14 Visual and mechanical inspection. Prior to assembly into the unit(s), assemblies, modules, and printed circuit boards shall be examined for conformance with the equipment drawings. All external moveable controls shall be operated mechanically. Assembled units shall be examined for conformance with equipment drawings, workmanship, and marking requirements. The equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify conformance with those portions of 3.9 which can be determined visually.

4.15 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 4Q-1-1-ØØ-00-NR-X-ED-Ø-00-A

5.1.2 Level B. 3Q-1-1-ØØ-00-NR-X-ED-Ø-00-B

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The X-ray probe is intended to be used to provide information to a unit commander as to the nuclear radiation present in the area over which the equipment is deployed. This radiation, which may have been produced by a nuclear explosion, will be measured as X-rays of Pu-239 from the fallout of the explosion. The equipment is intended to be used as portable survey instruments.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or Level B preservation and packing (see Section 5).

c. When rough handling test is required (see Section 4).

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of first articles to be submitted for approval (see 3.1).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted and replaced by a letter identifying the particular design for example DT-590W/PDR-56F. The contractor should apply for nomenclature in accordance with applicable clause in the contract (see 1.1).

Custodian:
Army - ER

Preparing Activity:
Army - ER
(Project 6665-A420)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Electronics Research &
Development Command
ATTN: DELCS-PE
Fort Monmouth, New Jersey 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER *(Number)* AND TITLE

PROBE, RADIAC DT-590( )/PDR-56F MIL-P-49392(ER)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID RESTRICTIVE LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:04Z
Check the source to verify that this is the current version before use.